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ROUGE: A database of disaster impacts in the Global South using Red Cross reports and Large Language Models

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Title

ROUGE: A database of disaster impacts in the Global South using Red Cross reports and Large Language Models

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Main Text: ~ 7000 words

Abstract: 170 words

3 Tables

6 Figures

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Abstract

High quality data on natural hazard damages are crucial for effective disaster risk management. Yet, existing impact datasets remain limited and often biased toward Northern countries and monetary losses. To help address these gaps, we present ROUGE (Redcross Operations Unified Global Emergency database); a new socio-economic impact database obtained using textual operational reports from the International Federation of Red Cross and Red Crescent Societies (IFRC). These reports are systematically collected and provide broad coverage of regions that are commonly underrepresented in existing impact datasets. Using large language models (LLM), we extract qualitative and quantitative information on a wide range of non-monetary impacts at national and sub-national scales. The resulting dataset documents socio-economic impacts of natural hazards on the population, infrastructure and economy with a spatial detail reaching the subregional level. This resource is designed to support research and applications that require geographically explicit information on socio-economic impacts of disasters, enabling more precise and inclusive analyses of socio-economic consequences of natural hazards worldwide.

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Background & Summary

Natural hazards exact a heavy toll on society, and have caused over 2 million fatalities and 3.6 trillion Euros in damages worldwide in the past 50 years [1]. This trend is expected to continue in the future, as many disasters become more frequent and severe due to climate change and exposure shifts [2].

Within this context, understanding how natural hazards lead to impacts is crucial for improving risk management and reducing damage. Gaining such understanding demands a comprehensive view of where and when disasters occur, as well as the magnitude of their impacts. While several databases recording the impacts of natural hazards at the global [1] or regional scale[3] do exist, they suffer from geographical biases [4], which can lead to under-reporting for the Global South [5] and result in overlooking certain risks in the most vulnerable regions. A second gap is that most impact databases focus on monetary losses, disregarding human, social, and environmental losses [6]. This focus undermines the importance of non-monetary, intangible losses and further amplifies the bias towards impacts in the Global North,

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where more financial and insured assets are concentrated. Finally, only a few databases report impacts at the subnational scale[7]. This lack of spatial detail constraints impact modelling precision, especially because hazard information is available at finer resolutions than the national scale.

With the growth of digital data repositories and recent advances in artificial intelligence (AI), a larger volume of data has become exploitable. AI-based methods are increasingly being used to improve the understanding and characterisation of natural hazards[8] and their impacts[9–11]. In particular, natural language processing has demonstrated strong potential for extracting information from scattered and unstructured data sources, where manual extraction is too time-demanding. For instance, recent work has shown how natural language processing can be used to automatically identify drought impacts in large collections of news articles, extracting details about the affected sectors, locations, and reported consequences through rule-based and machine learning classifiers [9]. However, the performance of keyword search and rule-based approaches is limited, requiring highly detailed and context-specific dictionaries[12]. Moreover, traditional classification models require large volumes of annotated training data, which are time-intensive to produce, especially for rare impact types [9]. In such a context, large language models (LLM) represent a more flexible and extensive tool for data extraction. Latest studies demonstrate the effectiveness of LLM-based approaches in systematically retrieving and analysing impact information [11,13,14].

Despite advances, the use of LLMs for extracting disaster and risk information from text data is still at an early stage. A recent systematic review highlights that challenges remain in handling temporal and spatial biases, and selecting appropriate modeling and post-processing strategies[15]. In particular, validating the extracted information is crucial, given LLMs' propensity to produce inaccurate or unsupported information[16]. This validation task remains time-consuming, especially when reference data is scarce and the data extracted is complex.

Against this background, the International Federation of Red Cross and Red Crescent Societies (IFRC) reports impacts from natural hazards since 1919[17] in the form of written operation reports. Part of the data that can be found in these reports is available in machine-readable format through the IFRC GO platform[17] and the IFRC's Montandon initiative[18]. These databases only include figures on the number of fatalities, people affected, injured, displaced, and missing at the national level. However, the reports provide information on a wider range of impacts, such as health and infrastructure damage, and at a finer spatial resolution. Hence, these written

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reports represent a valuable and largely underexplored resource for creating impact datasets[15]. Another advantage of this data is its explicit focus on disasters, which reduces the biases that often affect other impact datasets derived from news[8], or Wikipedia articles[19]. Moreover, the IFRC provides systematic reporting on events affecting Global South countries, which are frequently underrepresented in global impact datasets. As such, the IFRC reports can provide a more inclusive and balanced perspective on disaster impacts.

The combination of societal urgency, new artificial intelligence tools, and increasing volumes of text data provides a unique window of opportunity for studying the impacts of climate extremes. In this study, we leverage data from IFRC's textual reports and LLMs to produce a new socio-economic impact database: ROUGE (Redcross Operations Unified Global Emergency database). Specifically, we focus on unconventional, non-monetary, qualitative, and quantitative impacts at the national and sub-national levels. Our database records the impacts of 776 disasters in 145 countries for 14 different hazards from 2016 to 2025. The database covers impacts for 20 subtypes of impacts from three main types of impacts, including *Human*, *Infrastructure and Service Access*, and *Economy and Culture*. We expect this database to help the investigation of disasters and their risks, by providing new data on natural hazards and their impacts in regions which suffer from under-reporting in conventional databases.

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Methods

The extraction and processing of text data from IFRC reports followed four main steps (Figure 1). First, we scrap the reports and convert them to text format. Second, we extract and clean the text for analysis (Section Preprocessing). Third, we prompt an LLM to identify and extract impact data from the cleaned text (Section Data extraction). Finally, we standardise the extracted outputs, remove inconsistent elements and geolocate the reported locations (Section Post-processing). For technical validation (Section Technical Validation), we evaluate the uncertainties and its propagation throughout the pipeline (Section Quality flags analysis), validate the extracted information against manually labelled data (Section Validation with manually labelled data) and compare our database with established impact databases (Section Comparison with external data).

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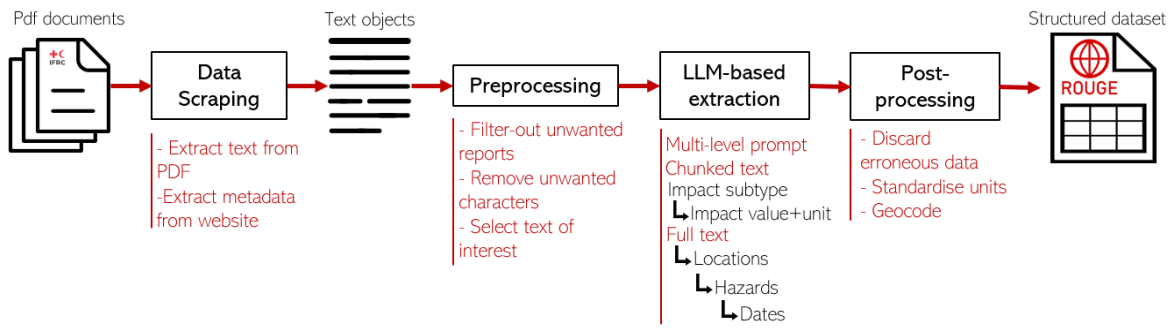


Figure 1 : Flowchart of the impact extraction pipeline.

Data

We use operational reports from the Disaster Response Emergency Fund (DREF) and the Emergency Appeal Fund of the IFRC as source data to extract the impacts of natural hazards. The reports are issued by National Societies of the IFRC in cases of humanitarian crises or disasters, covering small-scale crises (relatively narrow geographical scope and a with a low number of affected people) to large-scale disasters (a large area of a country or countries is concerned and a very large number of people are affected)[20]. These reports describe the extent of the crisis or disaster, including the area and populations affected, as well as the drivers that lead to the emergency situation (including natural hazards, epidemics, and conflicts). The reports also include detailed descriptions of the impacts on the population, the infrastructure and the environment, as well as the planned response measures for disaster relief. The IFRC identifies events using *appealCode* that uniquely identify each crisis. For each *appealCode*, the IFRC can issue several reports that provide updates on the operation's status, including a mandatory final closing report when the emergency appeal has been terminated[21].

We only consider reports from 2016 to ensure structural consistency throughout the reports. This corresponds to 2610 reports over 1221 unique appeals. To avoid extracting duplicate impacts for the same event, we select a single report per appeal. Specifically, we keep the latest available report for each event, as it provides the most up-to-date quantification of the impacts. Next, we select reports for which the associated disaster type in the IFRC metadata corresponds to one of the following natural hazards: Drought, Flood, Glacial lake outburst, Cyclone, Hurricane, Typhoon, Storm, Tornado, Heatwave, Coldwave, Mass movement, Earthquake, Volcano, Tidal Wave, and Wildfire. For reports without any disaster type, we check whether any of these hazard terms are mentioned in the report title or first 200 characters of the

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text using a keyword search. This further removes 471 reports and reduces the number of unique appeals to 781.

We scrape the reports from the IFRC website[17] and download them as PDFs. The text is extracted using the pdfplumber Python package[22]. All images are discarded, and only text, including tables, is retained. The text extraction failed due to a corrupted or unexploitable file for five of the reports, resulting in a final set of 776 unique appeals.

For comparison(see Section Comparison with external data), we used the EM-DAT[1] and IFRC GO[17,23] impact datasets. EM-DAT is a widely used impact dataset provided by the Centre for Research on the Epidemiology of Disasters[1]. It contains information on more than 27000 events worldwide from 1900 to the present day. The dataset gathers information from various sources such as UN agencies, non-governmental organisations, reinsurance companies, research institutes, and press agencies. It targets major events that resulted in at least 10 deaths and/or 100 affected people and/or were declared as an emergency by the state and/or were associated with an international assistance call. Precise information about the impacted location can be provided, but the impact is only available at the country level. We also considered the IFRC GO database, which compiles hazard events and associated impact data from IFRC DREF and Emergency Appeals reports for five categories of impacts (number of fatalities, people affected, injured, displaced, and missing), at the national level.

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Preprocessing

To ensure machine-readability and facilitate the use of LLMs, we conduct several preprocessing steps on the reports´ raw text using the Spacy python library[24]. First, we identify the language of each report using a language detection pipeline[25]. Results show that only 3 reports are not written in English. These reports are discarded to avoid needing to adapt our pipeline to other languages. However this does not result in any loss of unique appeal. Next, we remove unwanted text and characters such as URLs, ligatured characters, and multiple spaces. Given the relatively structured format of the reports, we extract only the sections that contain information on the hazard and its impacts. The section selection is done using section headers and keywords identified during review of the reports. Headers and keywords used for selection can be found in the appendix (Section Report sections selection). When no target text is found for a report, the report is discarded. We validate this selection process by reviewing the selected text of the 39 reports used for internal validation and we also check that the discarded reports

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did not contain useful data. This selection reduces the text we process, from an average of 223 sentences per report to 70 sentences, improving efficiency and the relevance of the information provided to the LLM.

Impact data typology and database structure

We define a socio-economic impact typology comprising 3 impact types and 20 subtypes (Table 1). The typology is developed using a combination of inductive and deductive approaches. First, we conduct a data-driven topic analysis of the reports to identify the impacts mostly frequently reported. Additional subtypes are also identified a posteriori when manually annotating the dataset. We then attempt to align these with the EM-DAT and UNDRR SENDAI frameworks[26] as indicated in Table 1. In some cases, the same level of granularity could not be achieved. For example, for *Residential buildings*, the Sendai Framework distinguishes between damaged and destroyed dwellings. However, because impact information is not reported consistently across these subcategories in IFRC reports, we extract impacts only for a broader *Residential Buildings* category that includes both destroyed and damaged buildings. When applicable, we extract information on the degree of damage using the extracted *impactUnit* (see *damageDegree* in Table 3). We also include categories that are not explicitly part of the Sendai Framework but are available in EM-DAT, such as *Affected People* and "*Homeless People*. In addition, the categories *Displaced People*, *Human Health and Wellbeing*, and *Informal Settlements* are introduced because these impacts are frequently mentioned or quantified in the reports. When available, we extract quantitative values associated with our impact typology, but we also retain qualitative mentions of impacts when no explicit numerical value is present in the target text. Besides the impacts, we also extract information such as the hazard type, location, and date. The hazard typology is adapted from EM-DAT hazard typology[1] and is described in Table 2. However, as the EM-DAT categories*Storms* and *Extreme Temperatures* are broad, we further disaggregate these hazards. *Storms* are further classified into *Tropical Storm*, *Convective Storm*, and *Other Storm*, and *Extreme Temperatures* into *Extreme warm temperature* and *Extreme cold temperature*. This refinement enhances the granularity and specificity of hazard attribution while preserving consistency with EM-DAT. We additionally include *Epidemics* and *Conflicts*, given their role in exacerbating disaster impacts and compounding vulnerability. Our processing framework only retains reports that specifically mention natural hazards. Hence, these two additional hazards are only considered in the context of compound events involving

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other natural hazards and are not treated as standalone hazard events. The corresponding mapping to the EM-DAT categories is also provided.

Impact Maintype	Impact Subtype	Description	UNDRR Sendai category
Human	Affected People	Total number of individuals impacted.	
	Injured People	Number of people injured, including those hospitalised or admitted.	B-2
	Displaced People	Number of people forcefully displaced or evacuated before or following the event.	
	Homeless People	Number of people losing housing following the event.	
	Missing People	Number of people unaccounted for following the event.	A-3
	Human Deaths	Number of fatalities.	A-2
	Infected and Ill People	Number of people contaminated (cases) by an infectious disease.	B-2
	Human Health and Wellbeing	Generic impacts on human health (physical, mental) or wellbeing not necessarily associated with the spread of an infectious disease.	
Infrastructure and Service access	Transportation	- Roads and transportation infrastructure (e.g. bridges, highways..) impacted. - People losing the capacity to move safely and efficiently between locations.	D-4; D-8
	Water, Sanitation, and Hygiene	- Number of water, sanitation, and hygiene infrastructure such as sewage networks, drainage systems, wastewater treatment plants, etc. impacted. - People losing safe, clean, and consistent supply of water for drinking, sanitation, hygiene services and other essential uses.	D-4; D-8
	Healthcare	- Number of healthcare infrastructure such as hospitals, healthcare centers, pharmacies, clinics, etc. impacted. - People losing the ability to obtain needed medical services, including preventative care, emergency services, etc.	D-1; D-7
	IT and Communication	- Number of IT and communication infrastructure such as data centers, communication towers, and cables impacted. - People losing communication access.	D-4; D-8

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	Residential Buildings	- Number of residential buildings (e.g. houses) impacted.	B-3; B-4;
	Informal settlements	- Number of informal settlements such as refugee camps, slums, tents, etc. impacted.	
	Education	- Number of education infrastructures such as schools, universities, etc. impacted. - People losing the ability to attend educational institutions and receive instruction.	D-3; D-6
	Power and Energy Production	- Number of energy production infrastructures such as power plants, turbines, grids, pipelines, etc., impacted. - People losing access to electricity and services related to power production.	D-4; D-8
	Agriculture and Access to Food	- Number of agricultural infrastructures such as farms, warehouses, greenhouses, fisheries, etc. impacted. - Number of crops, agricultural production and forest impacted. - Total number of animals impacted, including terrestrial and aquatic species (e.g. perished animals or fishes). - People losing access to a secure food supply.	C-2;
	Undefined Infrastructure and Service Access	- Any identified impact on infrastructure where the type of infrastructure impacted is not clearly defined, e.g. critical infrastructure, public infrastructure, access to basic services.	
Economy and Culture	Recreation, Tourism, and Culture	- Tourist attractions and cultural sites impacted.	C-6;
	Economic and Livelihood	- Any identified impact on the economy or living conditions.	B-5; C-3; C-4;

Table 1**.** Impact types used in this study, description and correspondence to UNDRR Sendai categories.

Hazard	Description	EM-DAT main type
Drought	Prolonged lack of precipitation.	Drought
Wildfire	Uncontrolled natural fires.	Wildfire
Earthquake	Sudden tectonic shifting. Also includes tsunami.	Earthquake

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Mass movement	Any type of downslope movement of earth materials. Also includes Landslides and rockfalls.	Mass movement
Volcanic activity	Eruptions and related phenomena.	Volcanic activity
Flood	River, coastal, flash and ice jam flooding.	Flood
Wave action	Wind-generated surface waves (e.g. Rogue waves, seiche).	Wave action
Extreme warm temperature	Prolonged, abnormally high heat.	Extreme temperature
Extreme cold temperature	Prolonged, abnormally low cold.	Extreme temperature
Tropical storm	Tropical cyclonic storms (e.g tropical cyclone, tropical storm, hurricanes, typhoons).	Storm
Convective storm	Convective-related storms (e.g. severe convective storms, derechos, hail, tornado).	Storm
Other storm	Any type of storm which does not correspond to a tropical cyclone or convective storm (e.g. extra-tropical storm, snow storm).	Storm
Epidemic	Sudden outbreak of disease.	Epidemic
Conflicts	Disagreements or disputes between different groups, organizations, or states.	

Table 2. Hazard typology used in this study, descriptions, and correspondence to EM-DAT hazard main type.

Data extraction

We use the model meta-llama/llama-4-scout-17b-16e-instruct [27], to identify and extract natural hazard impacts. Among the models evaluated, it provides the best balance between performance and computational cost (see Supplementary Sensitivity analysis). Model inference is done through the GROQ cloud platform[23].

The extraction of the impact data is done per report using a sequence of prompts[29] applied to either text chunks or the full text. First, the input text is split into chunks of sentences (approximately 1,000 characters) which are used for impact identification and extraction. This chunk-based extraction ensures that the LLM captures most of the relevant impacts without being constrained by its context window. Second, the full report text is used to retrieve contextual information for each identified impact. This staged approach allows us to better guide the extraction by providing the LLM with more focused information at each extraction step, while

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maintaining control over the outputs and limiting the length of the output context windows. The extraction process over the chunks starts with the following prompt:

Identify subtypes of impacts from the text, given a pre-defined list of impact subtypes and their definitions. The output is the list of identified subtypes.

> *For each identified subtype of impacts, extract the values and units of those impacts*. Extracting all impacts at once prevents the reuse of the same values for different identified subtypes. The output is a list of dictionaries.

For each identified impacts, the extraction continues as follows, using the full report for context:

For each identified impact (defined as a subtype, a value, and a unit), identify the affected locations. We allow a single impact to have multiple affected locations to account for aggregated reports of impacts. The output is a dictionary.

> *For each identified impact (defined as a subtype, a value, a unit, and a location), identify the hazards driving the impacts.* We allow for multiple hazards to account for compound events. The output is a dictionary.
>
> > *For each identified impact (defined as a subtype, a value, a unit, a location, and hazards), identify the starting and ending dates*.

At each step, we use the pydantic library[30] to validate the output. This validation allows us to control that each desired field is (i) present in the extracted json, (ii) of the correct data type e.g. string, integer, and (iii) contains only allowed values when value restrictions apply.

The validation on the value constraints is applied to the *impactSubtype* and *hazards* fields. If validation fails, we reprompt the LLM once with the validation error so it can correct the invalid output. Furthermore, we purposely extract unwanted “control” subtypes (see Supplementary Control impact subtype), such as the money raised by the appeal or the number of assisted people, as this prevents these figures from being wrongly extracted and classified as other similar impact subtypes, such as *Economy and Livelihood* or *Affected People*. These control subtypes are later discarded from the database (see Section Final data filtering). To further constrain the LLM during extraction, some impact subtypes (see Supplementary Impact Subtypes - Division) are temporarily divided into more specific categories before being aggregated back into their original subtypes during post-processing. To avoid duplicates, qualitative impacts sharing the same subtype are grouped, and contextual information (locations, dates, and hazards) is extracted once per group.

To ensure traceability and for validation purposes, we also query the LLM to provide the exact text excerpt used at each step for data extraction.

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Post-processing

We conduct several post-processing steps to ensure the consistency of the extracted data and standardize the final database. First, the extracted data is harmonized and standardized using regular expression and natural language processing (Section Harmonization and standardization). Next, locations are assigned geographical coordinates through a geocoding algorithm (Section Geocoding). Finally, error tracking procedures and consistency checks (Section Quality flags and issue tracking) are conducted for validation and final filtering of the data (Section Final data filtering).

Harmonization and standardization

First, for quantitative impacts, the *impactValue* is evaluated to ensure that it falls within the defined bounds, such that $impactValueMin \leq impactValue \leq impactValueMax$. We then reclassify and standardise the impact subtypes and hazard types using regular expressions. We format the *impactUnit* with the following four sub-steps; (i) numerical expressions detected in the unit field are moved to the *impactValue* field (e.g., "*10 thousand people"* becomes an impactValue of *10'000 and an ImpactUnit of people*), (ii) monetary units are converted to EUR using regular expressions and a currency converter[31] package, (iii) metric units, including distance, mass, surface, and volume, are standardized to SI units, and (iv) "measured" units are standardized using regular expressions and pre-defined conversion factors (e.g. *4 households* becomes *12 people*, *2 maternities* becomes *2 healthcare structures*).

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Geocoding

We convert textual location descriptions into georeferenced polygons. This step is done not only to check the correctness of the location identified by the LLM but also to enrich it with geographic boundaries.

For administrative boundaries, we rely on the *GAUL*[32] dataset for ADMIN1 and ADMIN2 and on Natural Earth[33] for ADMIN0. In this context, ADMIN0 refers to the country level, ADMIN1 refers to the first subnational administrative level, such as states, provinces, or regions, and ADMIN2 refers to the second subnational administrative level, such as districts, counties, or municipalities, depending on the country. Such datasets provide Polygons or Multi-Polygons geometries for all countries. However, because administrative levels beyond ADMIN2 (e.g.

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subdistricts, villages, neighbourhoods) are not consistently available across countries, we restrict our analysis to level 2 (ADMIN0, ADMIN1, and ADMIN2).

The textual locations are first converted to geographic information using the OpenStreetMap (OSM) Nominatim tool[34]. This service returns an OSM address with a geometry in the form of a Point, Polygon, or MultiPolygon object for a given set of location and country. An OSM address may include multiple address details (also known as *address keys* [35]), such as city, suburb, hamlet, street, and district.

To convert textual location strings into polygons, we first identify the most likely location name and administrative level using a fuzzy matching procedure based on OSM address keys. The corrected location is then matched to the corresponding Natural Earth or GAUL polygon. When no direct match is found, a series of fallback strategies; including alternative OSM address keys, multilingual translations, spatial matching, and using progressively coarser administrative levels; is applied to maximize the similarity between the location identified by the LLM and the available references administrative units. A detailed description of the fallback procedures is provided in the Supplementary Geocoding Process.

This process yields a polygon for each textual location. Because reported impacts may reference multiple locations at different administrative levels, the polygons of each resolution identified are kept in the final geometry.

Quality flags and issue tracking

We introduce a system of quality flags to monitor the outcomes of the different extraction and post-processing steps. The flags allow tracking the different processing steps that have been undertaken for each database entry, and the possible inconsistencies or errors that might have occurred. The flags can be grouped into different categories: *Extraction error* - an error occurred during the LLM extraction, *Post-processing* - a specific post-processing step has been undertaken, *Post-processing error* - an error has occurred during a specific post-processing step, and *Controlled removals* - an entry has been identified as undesirable and must be removed from the final database. The *Controlled removals* flags track control subtypes, unwanted units or unwanted hazards.

These flags are complemented with several consistency checks, which are referred to as *Warnings*. The consistency checks consist, for instance, in checking that the extracted impact

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value can be found in the original text or that a date field has the correct format. Finally, we identify critical flags which must be filtered out from the final database as *Error removals* flags. These flags correspond to entries for which one of the following occurred: the impact subtype or impact unit could not be identified or standardized, a number of people is higher than the country's population, the country and locations are both missing, both start and end years are missing, the hazards could not be identified.
A complete description of the flags can be found in the database structure table (Table 3). Their interpretation regarding the quality of the final database is detailed in the Section Quality flags analysis.

Final data filtering

Finally, we perform the following cleaning steps. We (i) remove duplicated entries, (ii) filter out entries for which at least one of the *Controlled removals* flags has been raised, (iii) filter out entries for which at least one of the *Error removals* flags has been raised.

Field category	Field name	Data type	Description
Data	impactType	str	Impact type from predefined list
	impactSubtype	str	Impact subtype from predefined list
	impactValue	float or None	Numerical value quantifying the impact
	impactValueMin	float or None	Lower bound estimate of the impactValue
	impactValueMax	float or None	Upper bound estimate of the impactValue
	impactUnit	str	Unit of impactValue
	damageDegree	str	Degree of damage. Either "partially damaged", "fully destroyed" or "unspecified"
	startYear	int or None	Start year of impact
	startMonth	int or None	Start month of impact
	startDay	int or None	Start day of impact
	endYear	int or None	End year of impact

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	endMonth	int or None	End month of impact
	endDay	int or None	End day of impact
	hazards	list	List of hazards associated with the described impact
	location	list	List of locations associated with the described impact
	locationLowestAdmin	str	Lowest administrative level identified with the geocoding
	locationPolygon	str	Name of the location polygon identified with the geocoding
	geometry	Polygon or MutiPolygon	MultiPolygon of the locations identified with the geocoding
	country_iso3	list	List of iso3 codes of the country identified with the geocoding
	continent	list	List of continents identified with the geocoding
	impactValuePrecision	str	Flag describing whether the impactValue is exact or approximate
Metadata	appealCode	str	Appeal code of IFRC report
	reportDate	str	Date of the report in YYYY/MM/DD format
	reportLink	str	Url link of the pdf report
	disasterType_IFRC	str	Disaster type identified by the IFRC
	sourceExcerpts	str	Text excerpts used for the information extraction
Extraction error flags	valid_errors_impactValue	int	Validation error impactValue extraction. Count of validation errors during the extraction of the impactValue.
	valid_errors_loc	int	Validation error location extraction. Count of validation errors during the extraction of the location.
	valid_errors_haz	int	Validation error hazards extraction. Count of validation errors during the extraction of the hazards.
	valid_errors_dates	int	Validation error dates extraction. Count of validation errors during the extraction of the dates.

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Post-processing flags	flag_impactSubtype_reclass	bool	Inferred impactSubtype must be in the allowed list. True if extracted impactSubtype was not in the allowed list and has been reclassified via regular expression
	flag_reclass_subtype_from_unit	bool	True if the extracted impactSubtype has been reclassified based on the impactUnit
	flag_impactSubtype_merged	bool	True if the impactSubtype has been merged into one of the merged categories
	flag_SI_unit_standardization	bool	Standardization of metric unit to SI. True if the extracted impactUnit has been standardized to a SI unit.
	flag_non-SI_unit_standardization	bool	Standardization of non-metric unit. True if a unit has been reclassified using regular expression, e.g. schools -> education structures, bridges -> roads
	flag_people_unit_normalization	bool	True if normalization of people unit has been applied
	flag_remove_number_unit	bool	True if a number has been identified from the extracted impactUnit and has been moved to the impactValue from the impactUnit (e.g. impactValue=10, impactUnit="milion people" => impactValue=1000000, impactUnit="people")
	flag_currency_conversion	bool	True if a monetary unit has been identified but has been failed to standardized to the default currency
	flag_non_currency_unit_conversion	bool	Non-currency unit conversion. True if a non-metric unit has been converted. These include families -> 3*people, households -> 3*people, communities -> 100*people, villages -> 1000*people
	flag_unit_harmonization	bool	Unit harmonization. True if any of the following standardization and harmonization step was applied to the entry: normalization of people units, standardization of metrics units to the SI system, standardization of non-metric units, removal of number in the unit
	flag_unit_conversion	bool	Unit conversion. True if any of the following conversion step was applied to the entry: conversion of monetary unit, conversion of non-monetary unit (e.g households -> 3 people)
	flag_hazards_reclass	bool	All inferred hazards must be in the allowed list. True if any of the extracted hazard was not in allowed list and has been reclassified via regular

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			expression
	flag_inferred_startYear	bool	True if the startYear has been inferred. Possible methods are most_frequent_startYear and infer_from_endYear
	flag_nomin_translate	bool	True if, for at least one location of the entry, reverse geocoding is used to translate the location to other languages
	flag_osm_polygon	bool	True if, for at least one location of the entry, the polygon from OpenStreetMap is used to retrieve the GAUL polygon
Post-processing error flags	flag_unit_harmonization_error	bool	True if an error occurred during any of the unit harmonization steps
	flag_unit_conversion_error		True if an error occurred during any of the unit conversion steps
	flag_fail_country_iso	bool	True if the initial country failed to be converted to iso3
	flag_nomin_no_result	bool	True if, for at least one location of the entry, nominatim didn't return any location for all the fallback strategies
	flag_nomin_sim_below_th	bool	True if, for at least one location of the entry, the best-matching Nominatim result has a similarity score below the threshold.
	flag_non-SI_unit_standardization_error	bool	True if an error occurred during the standardization of a non-metric unit
	flag_SI_unit_standardization_error	bool	True if an error occurred during the standardization of the impactUnit to a SI unit.
	flag_remove_number_unit_error	bool	True if an error occurred during the removal of numeric value in the impactUnit
	flag_currency_conversion_error	bool	True if an error occurred during the conversion of a currency unit
	flag_non_currency_unit_conversion_error	bool	True if an error occurred during the conversion of a non-currency unit
	flag_unit_processing_error	bool	True if there has been an issue in any of the following stages of the postprocessing of the impact unit: remove_number_unit, non-currency and currency unit_conversion, SI_unit_standardization, non-

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			SI_unit_standardization
	flag_failed_startYear_inference	bool	True if the inference of the startYear failed
Warning flags	flag_value_not_in_text	bool	ImpactValue should be in the original text. True if impactValue cannot be found in the original text. None if there is no impactValue
	flag_value_no_unit	bool	True if an impactValue has no unit
	flag_partial_unit	bool	True if the impactUnit lacks the measured unit (e.g. “hectares” instead of “hectares of crops”)
	flag_percent	bool	True if the impactUnit is a percentage
	flag_unit_nonstd	bool	True if the extracted impactUnit is not part of the “standard units”
	flag_hazards_unknown	bool	True if at least one of the hazards could not be identified as in the allowed list
	flag_inconsistent_year	bool	True if any of startYear or endYear is smaller than the minimum allowed year or larger than the maximum allowed year. None if both startYear and endYear are missing
	flag_inconsistent_month	bool	True if any of startMonth or endMonth is not an integer between 1 and 12
	flag_inconsistent_day	bool	True if any of startDay or endDay is not an integer between 1 and 31
	flag_missing_startYear	bool	True if the field startYear is missing
	flag_missing_endYear	bool	True if the field endYear is missing
	flag_missing_startMonth	bool	True if the field startMonth is missing
	flag_missing_endMonth	bool	True if the field endMonth is missing
	flag_missing_startDay	bool	True if the field startDay is missing
	flag_missing_endDay	bool	True if the field endDay is missing

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	flag_startYear_after_endYear	bool	True if the startYear is after the endYear. None if any of startYear or endYear is missing
	flag_years_missing	bool	True if both startYear and endYear are missing
	flag_no_location	bool	True if the location is an empty list
	flag_location_to_country	bool	True if, for at least one location of the entry, no polygon could be matched, the country polygon is used instead
Error removal flags	flag_incorrect_unit	bool	True if any of the following flag is True: "flag_partial_unit", "flag_unit_nonstd", "flag_percent"
	flag_country_location_missing	bool	True if both the country and the location fields are missing
	flag_years_missing_after_inference	bool	True if start year was missing and the start year inference failed
	flag_unknown_subtype	bool	True if impactSubtype reclassification could not identify the subtype as in the allowed list
	flag_all_hazards_unknown	bool	True if at all the hazards could not be identified as in the allowed list
	flag_pop_cntry	bool	ImpactValue with “people” unit should be smaller than country’s population. True if impactValue with people unit is higher than country population
Controlled removal flags	flag_remove_cat	bool	True if the impactSubtype is in the list of impactSubtypes to remove: "DREF Allocation & Funding requirements", "Targeted People", "Assisted People"
	flag_remove_unit	bool	True if the impactUnit is in the list of units to remove: "children", "women", "male", "female", "volunteers", "staff", "beneficiaries", "branches", "national societies", "trainers"
	flag_remove_hazard	bool	True if all the hazards are in the list of hazards to remove (Epidemics, conflict)

Table 3**.** Structure of the impact database.

Data Records

We assemble a database with the structure described in Table 3. Each entry of the database represents one impact record, specified by an impact subtype, a value, a unit, a location (geometry), a date, and a driver (associated hazards). Each impact record is also

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complemented with metadata (Table 3), informing on the source (i.e. the appeal code uniquely identifying an IFRC appeal, the report date, the sentences that were used for the extraction of the impacts), and several quality flags, which track issues and possible inconsistencies (see Section Quality flags and issue tracking). The data will be made accessible via Zenodo under the *Creative Commons Attribution 4.0 International* license. The database is available in two formats: as a global csv file without geometries and as Parquet files with geometries, available globally and split by continents. A notebook detailing how to open and access the data is available in the code archive, which can be downloaded anonymously by the reviewers at https://figshare.com/s/99da5ff800fc56ae10c0. The data can be downloaded anonymously by reviewers at the same link.

Data Overview

The LLM-based impact dataset is created from IFRC reports and primarily captures impact in the Global South, as illustrated in Figure 2. Overall, our database gathers 15147 records of natural hazard impacts, over 776 unique events identified. The regions most affected are Africa and Asia, with 5416 and 4903 impact records in the database (Figure 2), respectively, covering 269 and 270 unique events (Figure 3). Figure 3 shows that floods, along with tropical and convective storms, are the hazards most frequently associated with the extracted impacts. Additionally, mass movements and epidemics also contribute substantially to the overall recorded impacts. 56% of the impacts extracted are geocoded at the country level (ADMIN0), 31% at the first-order administrative division (ADMIN1), and 13% at the second-order administrative division (ADMIN2).

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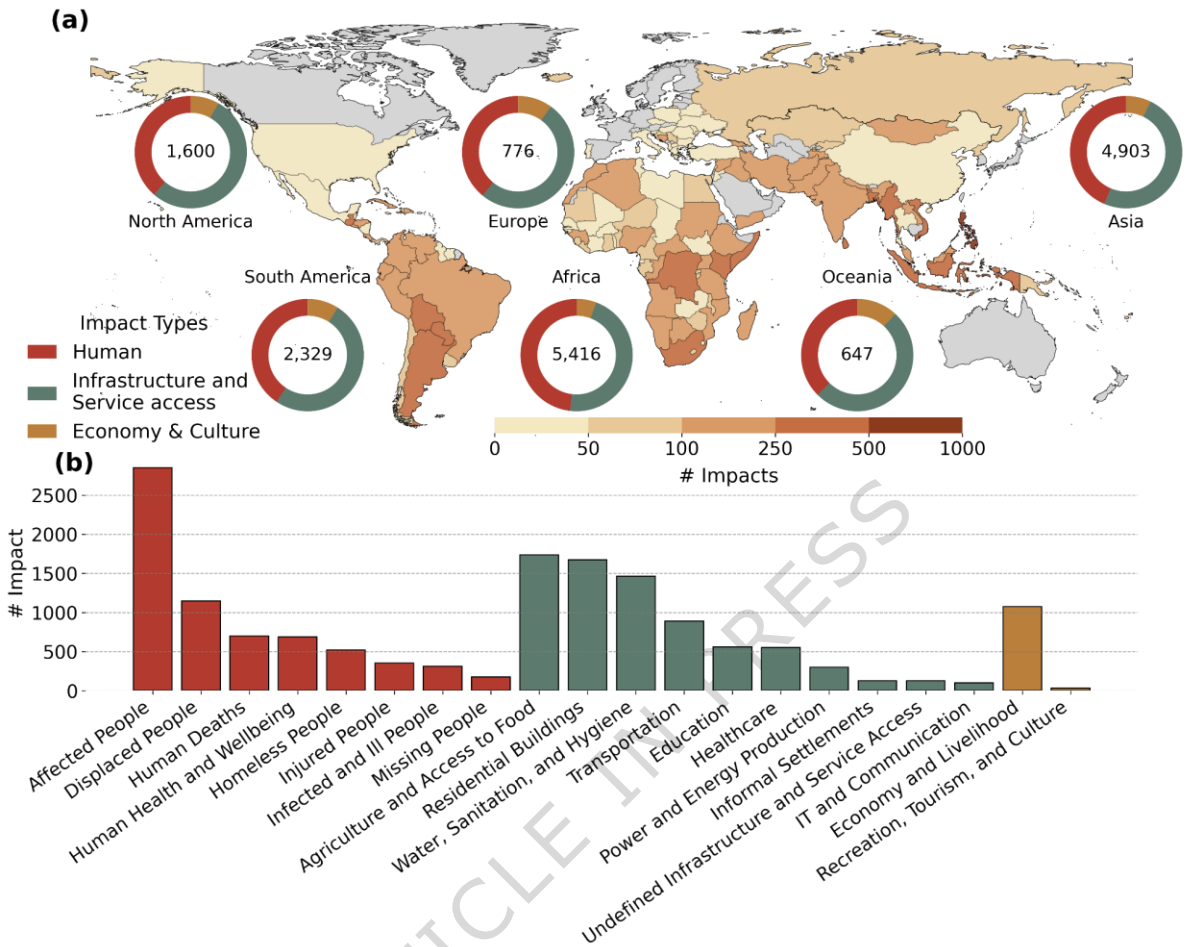
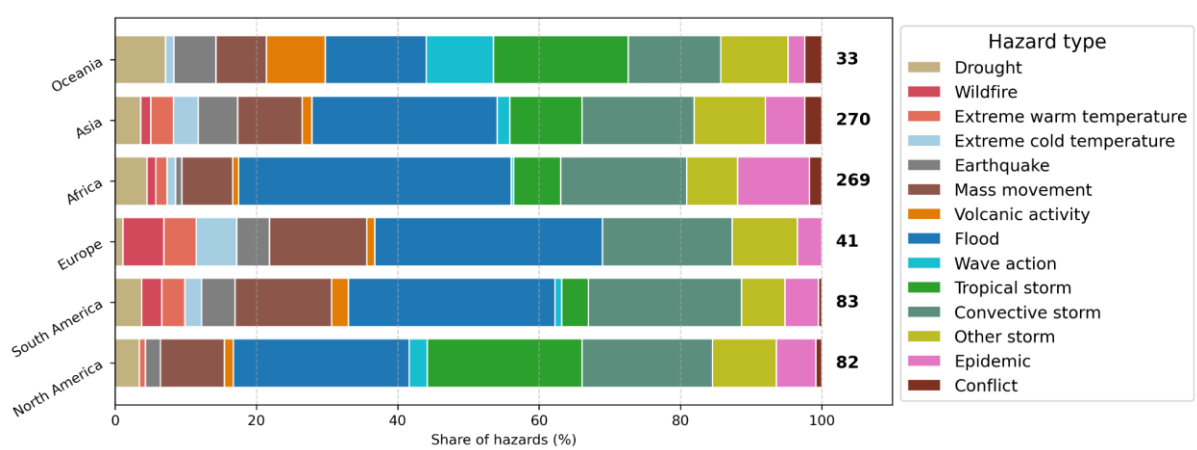


Figure 2: Upper panel (a) shows the spatial distribution of the number of impact records per country in colour. Circular bar plots show the share of *impactType* per continent, with the inner numbers indicating the total number of impact records. The Lower panel (b) is the histogram of the number of impacts in each *impactSubtype* category. Colours correspond to the *impactType.*



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Figure 3: Share of hazards (in %) to which impacts are associated, for each continent. Colours are ordered according to the type of hazard.

Technical Validation

Quality flags analysis

We analyse how possible inconsistencies and errors occurring at different stages of the pipeline distribute in the final database by using the flagging system described in Section Quality flags and issue tracking.

First, we examine the distribution of flags in the six different flag groups introduced; *Extraction errors, Postprocessing, Postprocessing errors, Warnings, Error removals,* and *Controlled removals*.

Figure 4a shows the distribution of the flags across these six categories. The color bars represent the number of database entries for which each flag is raised in the final database, with hatched portions indicating entries removed during the final filtering step (see Section Final data filtering). The flags discarded in the final filtering step are grouped in the categories *Error removals* and *Controlled removals* and marked in red.

The most represented flags in the final database are *Post-processing* flags affecting 13945 entries (92%) of the final database. This indicates the importance of the post-processing for the standardization and harmonization of the database. Most flags come from the unit harmonization step (*flag_unit_harmonization*), which tracks the standardization and harmonization of diverse units (e.g. “clinic” → “healthcare facility” or “3 households” → “9 people”, see Table 3). This highlights that a substantial number of database entries undergo a standardization procedure due to the heterogeneous and non-standardized reporting practices used in the original reports. The standardization and reclassification of the impact subtypes also represents a substantial part of the postprocessing, with about 38% of the entries of the final database being affected.

The *Warnings* flags are the second most represented category with 12979 flagged entries affecting 86% of the final database. This category is mainly driven by flags informing that the extraction of end dates has failed, with 59% of the entries having no end day, 51% no end month and 48% no end year. The database entries affected by these flags are kept in the final database as we consider that the impacts recorded in these entries are still exploitable besides the absence of end dates.

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Errors in the post-processing affect about 36% of the entries of the final database, and are mostly driven by the errors from the geocoding. These errors mainly arise from the inability of the geocoding procedure to match extracted locations with the corresponding reference administrative units. In such cases, the country is used as a fallback in place of the extracted location, resulting in the warning flag *flag_location_to_country* being raised for 39% of the final database entries. In comparison, errors occurring during the extraction step only affect 4% of the database entries. However, these flags mainly track errors in the formatting of the LLM output whereas most of the complex validation and standardization procedures occur during post-processing. Hence, the flags associated with the errors in the extraction steps only capture a subset of all the errors that may originate from the LLM-extraction.

Regarding the removal of database entries (*Error removals* and *Controlled removals* in Figure 4a), the majority of the removals originates from the controlled removals, which correspond to control units and categories (see Section Data extraction). These controlled removal led to the exclusion of 4686 entries of the initial database (21%) with 4289 entries specifically removed due to the control impact subtypes (*flag_remove_cat*). In comparison, 2695 entries of the initial database (12%) are removed due to uncontrolled errors, including 2649 database entries removed because of unidentified units (*flag_incorrect_unit*).

Figure 4b shows the distribution of flags grouped into the data variables the flags control for; impact subtype, value and unit, location, hazards, and dates. This allows us to highlight which components of the database are the most prone to errors or require the most post-processing or filtering. The distribution of the flag counts into these five categories confirms the complexity of accurately extracting and geolocating locations from text as the location stands out at the most represented category, affecting 73% of the final database entries. Next, flags associated with the validation and processing of the value and unit variables are the second most represented (65%). As highlighted in the previous paragraph, this is largely driven by the post-processing and harmonization of the units which are not reported in a standardized form. Flags associated with dates are raised for 63% of the final database entries and are led by the absence of end dates. We compare the number of missing date fields between the LLM-extracted and manually-labelled data (see Section Validation with manually labelled data) to determine if the failure to extract end dates is stemming from the LLM extraction or is caused by this information being absent from the original reports (not shown). We find that end days are missing for 50% of the entries in the manually labelled data, while end months and years are missing in 35% of the cases. While these numbers are slightly lower than for the LLM-extracted data, they confirm that a large part of the absences of end dates in the extracted data are driven by the initial data and

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not the extraction process. Finally, entries flagged because of the impact subtypes and hazards respectively represent only 38% and 1% of the final database, indicating that these fields are better constrained and easier to extract and process.

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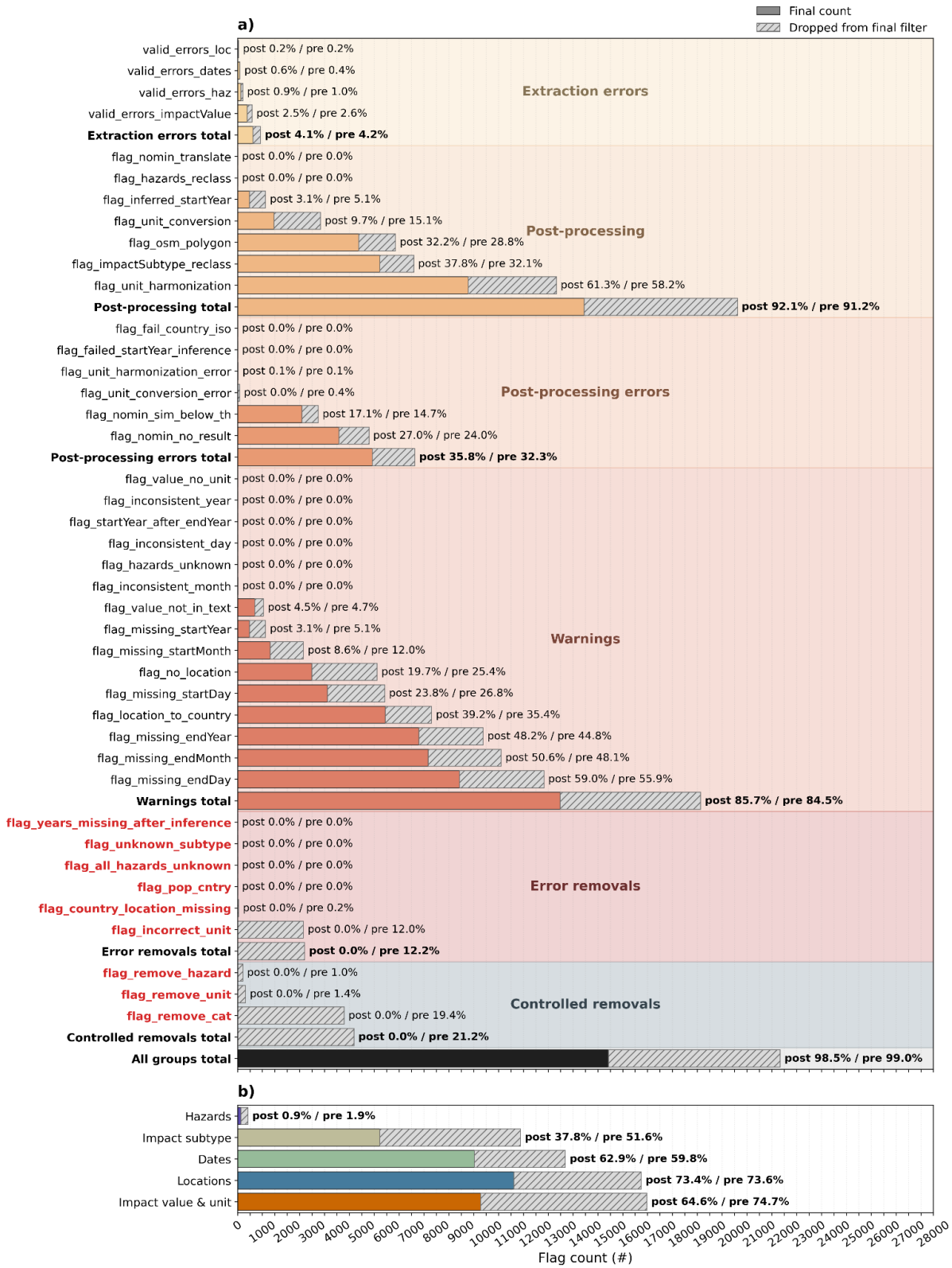


Figure 4 : Distribution of flags in the final database and before the final data filtering. Bars represent the number of database entries affected by each flag, colored according to the six flag groups described in Section Quality flags and issue tracking; *Extraction errors*, *Post-processing*,

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Post-processing errors, *Warnings*, *Error removals*, and *Controlled removals* (a), and according to the data variables the flag control for; *Hazards*, *Impact subtypes*, *Dates*, *Locations*, and *Impact value & unit* (b). Hatched sections of the bars represent the counts of flags that are removed from the final database using the filter based on the flags highlighted in red. Flags labels and flag groups are described in Table 3. The numbers at the end of each bar indicate the percentage of database entries affected by each flag, relative to the total number of entries before (*pre*) and after (*post*) the final filtering (described in Section Final data filtering).

Validation with manually labelled data

We validate our database internally by comparing LLM-extracted impacts with manually extracted impacts from 39 selected reports. These reports are selected to cover all identified hazard types and include events from every continent (see Supplementary Manually labelled reports). Across the selected reports, we identify a median of 28 impact-related sentences per report (range: 8–199), resulting in a total of 1,433 manually annotated sentences. This results in a validation corpus of about 5% of the total sample size, in line with sample sizes commonly used in large-scale text-mining and information extraction studies, where manual annotation is inherently time-intensive and therefore typically performed on a representative subset of the available data[36,37]. Manual extraction followed strict, pre-defined guidelines (see Supplementary Labelling Guidelines), and was performed by the authors with cross-reviewing. The labelled data acts as a control and represents the reference standard to which the LLM-extracted data is expected to adhere. From the 39 reports selected, we manually identified 464 quantitative and 227 qualitative impacts, spanning all the considered 20 impact subtypes and 13 hazards. In comparison, the automatic extraction using the LLM extracted 495 quantitative and 243 qualitative impacts, spanning 20 impact subtypes and 14 hazards.

Comparing the LLM-extracted with the manually-labelled impacts allows us to derive similarity scores, as well as the *precision* and *recall* of the extraction. The similarity scores are defined as a function of the examined variable's type and range from 0 for perfect dissimilarity to 1 for perfect similarity. Similarity of categorical variables is expressed using cosine similarity [38], similarity of numerical variables (*impactValue*) with a normalised relative difference (equation 2 in the supplementary *Matching algorithm for internal validation*), and geometries with the Intersection-over-Union (equation 1 in the supplementary *Matching algorithm for internal validation*). Precision and recall represent, respectively, the percentage of correctly extracted data over the total extracted and the percentage of correctly extracted data over the total manually-labelled.

In order to compute the similarity scores for each extracted impact, we use an algorithm that matches each manually-labelled entry to the most similar entry from the LLM-extracted data for the same report (see Supplementary Matching algorithm for internal validation). Figure 5 shows the average similarity per matching variables, separated by quantitative and qualitative data. The *match similarity* is the weighted mean of the variables' similarities for the matched data. We explain the lower *match similarity* in the qualitative data by the lower similarity of the impact subtypes, with some qualitative impact subtypes being more abstract, which may cause the LLM to perform worse at classifying the data into the correct subtypes. The similarity scores for the remaining variables are generally high, with values above 80% for the impact unit, the countries' ISO3 codes, the start year, the impact values, and the hazards. The similarity of the dates is lower (around 50%). We attribute this lower performance to the complexity of clearly identifying event dates, which the human labellers also encountered.

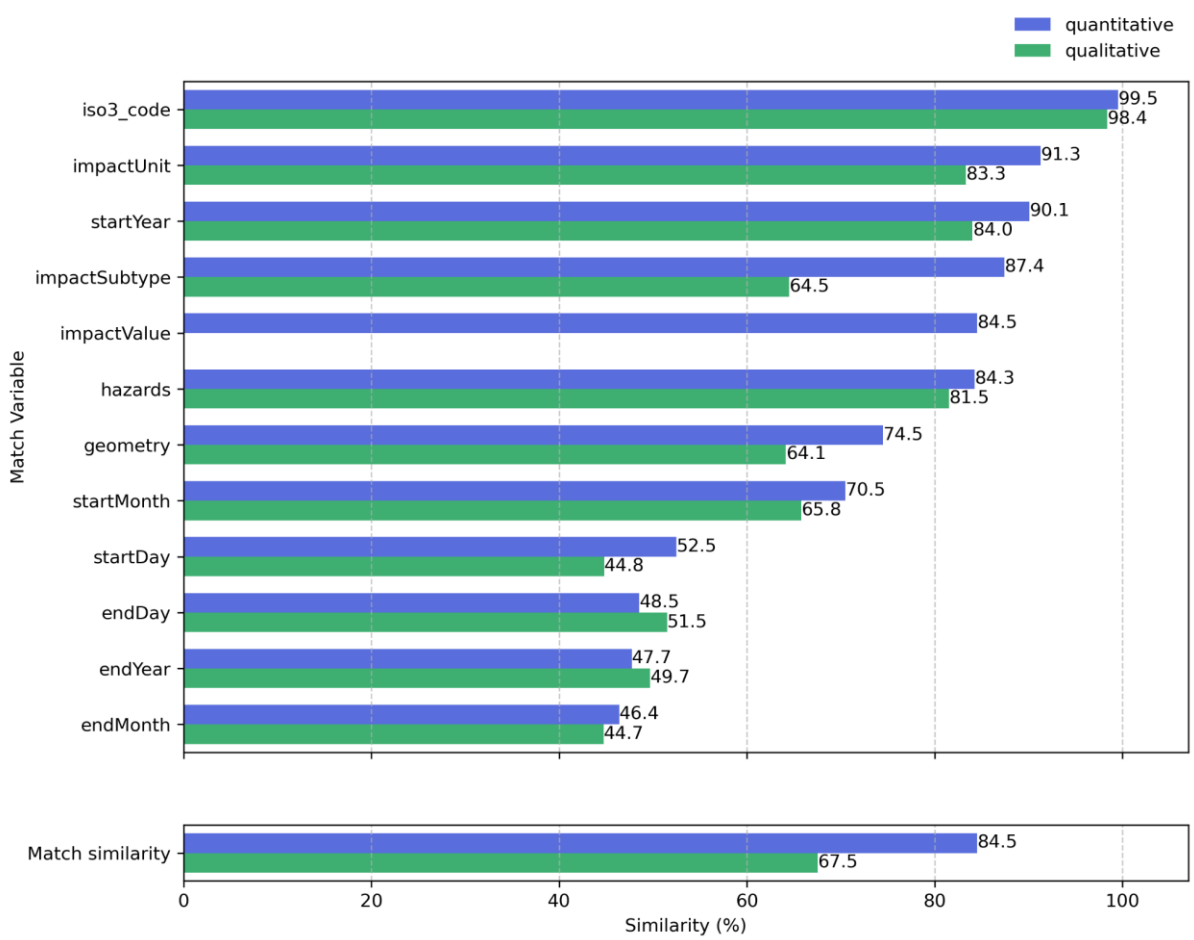


Figure 5: Similarity score per matching variable. The horizontal bars show the similarity score (%) for each of the variables considered for the matching of extracted and manually-labelled impacts. The bottom

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bar plot shows the average for all matches of the weighted mean of the similarity scores for all matching variables. Colours show results for the quantitative vs qualitative impacts.

Next, we compute the *Precision* and *Recall* using the matched data to estimate the predictive performance of our extraction:

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

where *TP* represents the True Positives, i.e. the number of LLM-extracted impacts for which we find a corresponding manually-extracted impacts, *FP* the False Positives, i.e. the number of LLM-extracted impacts for which we do not find a corresponding manually-extracted impact, and *FN* the False Negatives, i.e. the number of manually-extracted impacts for which we did not find a corresponding LLM-extracted impact. A perfect association should result in a *precision* and a *recall* score of 1. Lower *precision* indicates a greater tendency for the LLM to hallucinate, while lower recall indicates that the LLM misses a larger portion of the true impacts.

To quantify the sampling uncertainty, we approximate the 95% confidence intervals of the precision and recall metrics using a bootstrapping approach. For each sample from which the metrics are estimated, we sample with replacement 5000 random samples of true positives, false positives, and false negatives. We then compute the metrics from each random sample to obtain the bootstrapped distribution of the metrics. Finally, we use the 2.5th and 97.5th quantiles of the bootstrapped distribution as the bounds of the confidence interval.

Figure 6a shows *precision* and *recall*, and their estimated 95% confidence intervals for the total number of quantitative and qualitative impacts extracted. Overall, we find a precision of 0.71, and a recall of 0.75, resulting in a F1 score of 0.73 indicating that our approach is able to correctly extract about three quarters of the impacts. We observe a difference in precision between the extraction of quantitative and qualitative data, with quantitative data being extracted with a precision of 0.68 and qualitative data with a precision of 0.79, suggesting that the model performs better at inferring qualitative information of an impact over numerical quantification. Regarding recall, we obtain 0.73 for the quantitative data and 0.8 for the qualitative data. We explain the lower recall for the quantitative data by the difficulty for the model to comprehensively extract data from sentences where the information content is dense. This is the case for lists or tables which are formats in which numerical data is often reported. Figure 6b shows the precision and recall metrics per impact subtype. The 95% confidence intervals indicate that sampling uncertainty is high due to the smaller sample sizes when

grouping by impact subtype. For three of the 20 impact subtypes (*IT and Communication, Informal Settlements,* and *Recreation, Tourism, and Culture*), the sampling uncertainty is too large to exclude the precision and recall to be 0. For the majority of the remaining subtypes, their performance does not differ significantly from the average precision and recall. However, we note lower than average precision or recall scores for the categories *Healthcare, Human Health and Wellbeing, Infected and Ill People, Homeless People* and *Undefined Infrastructure and Service Access.* This lower performance can be explained by the fuzziness of these categories, leading to misclassification. For instance, the model will possibly assign a number of houses destroyed to the category *Homeless People* instead of *Residential Buildings*. Moreover, the model can classify duplicate mentions of the same impact inconsistently, leading to duplicate entries which are difficult to filter out.

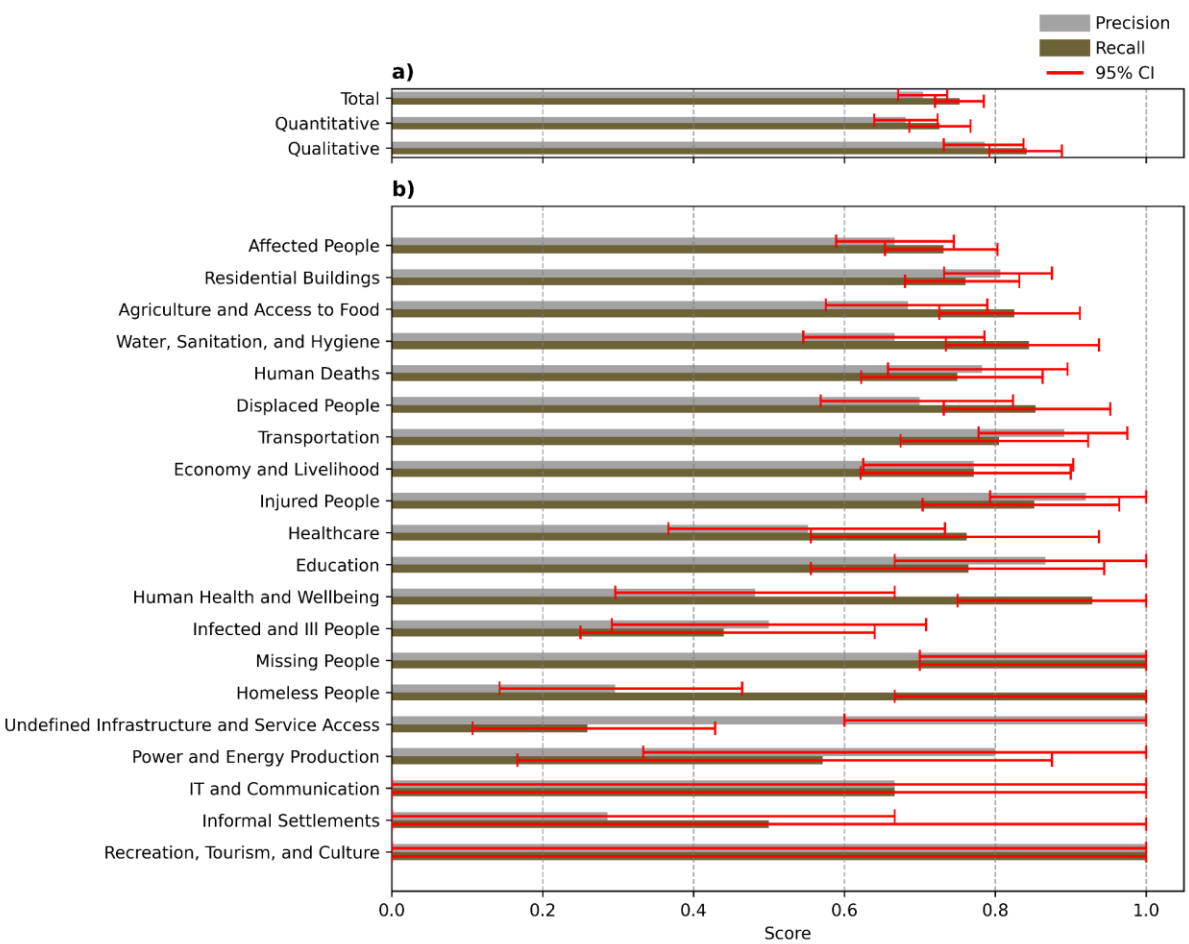


Figure 6 : Precision (light gray) and recall (dark gray) per quantitative vs. qualitative extraction (a), and per impact subtypes (b). Colorbars represent the value of the metrics estimated from the samples directly. Red error bars represent the 95% confidence interval estimated via bootstrapping.

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In addition, we run robustness checks to assess whether our approach is sensitive to the choice of the LLM used and if it is deterministic (see Supplementary Sensitivity analyses). Among the models evaluated, the selected LLM produces stable outputs and achieves the best overall performance.

Comparison with external data

We compare the quantitative impacts extracted by the LLM at the national level to EM-DAT and IFRC GO datasets. From our *impactSubtypes* (Table 1) we compare the categories *Affected People*, *Human Death* and *Homeless People* with EM-DAT and the categories *Affected people*, *Injured people*, *Displaced People*, *Human Deaths* and *Missing People* with IFRC GO. As EM-DAT uses the IFRC to source their records of the number of people affected in 65% of cases[1], we expect this category to match well across the three data sources.

To compare reported quantitative impacts, we match each event in ROUGE with corresponding events in EM-DAT and IFRC GO. Since EM-DAT and ROUGE do not share a common unique event identifier, we apply a stepwise matching procedure. First, each ROUGE report is linked to EM-DAT events occurring in the same year and country. When multiple hazards are listed in a given ROUGE report, we retain the hazard associated with the highest number of reported impacts and filter EM-DAT events to those whose *DisasterType* matches this dominant hazard. If multiple EM-DAT events remain after this filtering, we select the event with the entry date closest to the ROUGE report date. This procedure yields a one-to-one mapping between EM-DAT events and ROUGE events. For IFRC GO, matching with ROUGE is performed directly using the appeal code. We extract 688 events from IFRC GO over the 2016–2025 period, all associated with natural disasters, and successfully link 600 of them to the ROUGE-extracted events. Finally, as both EM-DAT and IFRC GO report impact at the country level, we aggregate our results at the ADMIN0. This is done level by retaining, for each record, the highest reported impact value at the coarsest administrative level. The proportion of reports containing quantitative impacts in ROUGE varies substantially across categories. For the 776 reports for which we extract quantitative impact data in ROUGE, quantitative information is available in 96% of the cases for the category *Affected People*, in 28% of the cases for *Injured People*, in 55% of the cases for *Displaced People*, in 54% of the cases for *Human Deaths*, in 16% of the cases for *Missing People*, and in 29% of the cases for *Homeless People*. Figure 7a compares the availability of quantitative information in ROUGE, EM-DAT, and IFRC Go. Using ROUGE as the reference dataset, we consider only the matched events for which ROUGE extracted a

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quantitative impact and assess whether the corresponding quantitative information is also available in EM-DAT and IFRC GO. To make comparisons consistent across categories, percentages are computed relative to the total number of reports in ROUGE containing quantitative impact, for each impact category. Across all impact categories, ROUGE provides quantitative information that cannot be found in the other databases. This pattern is particularly evident for the categories *Displaced People*, *Missing People*, and *Homeless People,* where quantitative information can be found solely in ROUGE for, respectively, 64%, 66%, and 92% of the matched events. For the categories *Affected People*, *Injured People*, and *Human Death*, quantitative information can be found across all databases. This is the case for approximately 30% of the matched events while quantitative information can be found in ROUGE and EM-DAT but not IFRC GO for an addition of 30% to 38% of the matched events, and in ROUGE and IFRC GO but not EM-DAT for an addition of 8% to 17% of the matched events. Finally, quantitative information for these three categories can be found solely in ROUGE for 19% to 26% of the matched events. These results highlight that coverage differs by impact category and by database, but they also indicate that ROUGE can complement existing databases by extracting quantitative impacts that are not systematically captured in EM-DAT or IFRC GO.

For events and impact subtypes where quantitative impact is available beyond ROUGE, we compare the reported impact values across the different sources. Jitter plots and boxplots in Figure 7b illustrate the distribution of the coefficient of variation (CV) defined, for each event i, and impact subtype s as the ratio of the standard deviation to the mean:

$$CV^{i,s} = \frac{\sigma({x^{i,s}}_{ROUGE}, {x^{i,s}}_{EM-DAT}, {x^{i,s}}_{IFRCGO})}{\mu({x^{i,s}}_{ROUGE}, {x^{i,s}}_{EM-DAT}, {x^{i,s}}_{IFRCGO})}$$

where ${x^{i,s}}_{d}$ is the quantitative impact value for the event i of impact subtype s from the database d, i.e. ROUGE, EM-DAT or IFRC GO. If a database d doesn't have any quantitative impact for a given impact subtype s and event i, ${x_d}^{i,s}$ is set as null and ignored for the calculations. A CV of 0 indicates perfect agreement among all databases, whereas higher values reflect increasing discrepancies between reported impact values. A CV of 1 indicates that the standard deviation equals the mean. Jitter points of Figure 7b are colored according to the combination of databases available for each comparison. Comparisons are performed either across all three databases when values are available from each source, or pairwise between ROUGE and either EM-DAT or IFRC GO when only two sources are present. Agreement levels vary substantially across impact categories. For the category *Affected People*, the median CV of

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0.47 indicates substantial divergence between databases, corresponding to differences on the order of 50% of the reported impact values. The highest CV values are observed either when all three databases are available (median CV = 0.67) or when comparisons involve only ROUGE and IFRC GO (median CV = 0.72), suggesting larger discrepancies associated with values reported in IFRC GO. In contrast, stronger agreement is observed between ROUGE and EM-DAT, with a substantially lower median CV of 0.16. For the *Injured People* and *Human Deaths* categories, better consistency is observed among the three databases, with median CV values of 0.17 and 0.30, respectively. In both cases, ROUGE shows stronger agreement with EM-DAT than with IFRC GO. For *Injured People*, the median CV is 0.0 for comparisons between ROUGE and EM-DAT, compared with 0.44 for comparisons between ROUGE and IFRC GO. Similarly, for *Human Deaths*, the median CV reaches 0.16 with EM-DAT and 0.19 with IFRC GO. For the category *Missing People*, relatively good agreement is found between ROUGE and IFRC GO, with a median CV of 0.26. By contrast, lower consistency is observed for the category *Displaced People*, a category available only in IFRC GO. Finally, relatively high CV values are observed for the *Homeless People* category. This could be explained by the potential lower extraction precision for this category as highlighted in Section Validation with manually labelled data. However, the limited number of reports with quantitative impact values in EM-DAT for this category should also be taken into account for assessing the robustness of this finding.

Possible explanations for the discrepancies in the values reported between the databases include differences in the specific report versions used in IFRC GO, limitations of the matching procedure applied to EM-DAT, or inaccuracies introduced during the extraction process in any of the sources, including the LLM-based pipeline, IFRC GO, or EM-DAT. These inconsistencies across data sources and impact categories further highlight the inherent challenges in reliably estimating and harmonizing quantitative impact information from natural disasters.

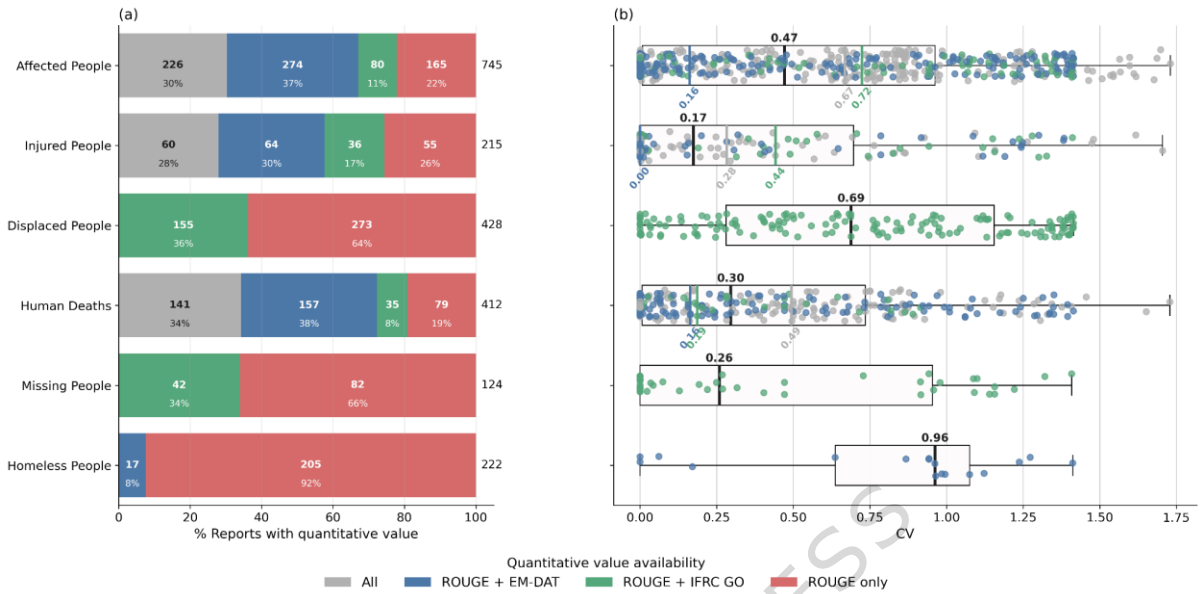


Figure 7: Color bars in (a) show the share of reports for which corresponding data is also available in external databases among the events for which quantitative data has already been identified in ROUGE Each line corresponds to an impact subtype: *Affected People*, *Injured People*, *Displaced People*, *Human Deaths*, *Missing People*, and *Homeless People*. Numbers on the right indicate the number of ROUGE reports per category with quantitative data. Colors distinguish data availability across sources: grey indicates overlap between ROUGE, EM-DAT, and IFRC GO; blue indicates overlap between ROUGE and EM-DAT only; green indicates overlap between ROUGE and IFRC GO only; and red indicates quantitative data available only in ROUGE. Numbers and percentages within each segment report the exact counts and shares relative to the ROUGE totals.
Boxplots in (b) show the distribution of the coefficient of variation (CV), with overlaid jittered points representing individual events with quantitative impacts available in ROUGE and at least one additional source. Each point corresponds to one event, and colors indicate the external datasets used for comparison. Vertical lines indicate the median CV computed across all observations (black) and stratified by data source combinations (grey, green, and blue).

Usage Notes

The data is available in csv format without the geometries and in Parquet format with the geometries, for the entire globe or separated by continent.

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The extraction of both quantitative and qualitative impacts using exclusively LLMs presents limitations, which are discussed in the Technical Validation section. In light of these limitations, we propose the following recommendations.

In cases of suspicious extracted information in any category (e.g., impactValue, impactUnit, location, dates), we encourage users to verify the corresponding sourceExcerpts, which contain the unprocessed sentences used by the LLM for impact extraction.

During the internal validation, we also identified potential duplicated impacts. Therefore, for analyses involving the same locations or impactTypes, we recommend using maximum values rather than summed values.

Users also have the possibility to operate further filtering using the quality flags (Table 3 and section Quality flag analysis). For instance, users can filter out entries for which event dates were only partly extracted.This enables tailoring the database to the users' needs.

Data Availability

The data will be made accessible via Figshare in csv format without the geometries and in Parquet format with the geometries, under the *Creative Commons Attribution 4.0 International* license. The data can be downloaded anonymously by reviewers at: https://figshare.com/s/99da5ff800fc56ae10c0

Code Availability

The code to produce the database, perform the technical validation and make the analysis plots is available anonymously at: https://figshare.com/s/99da5ff800fc56ae10c0.

Author Contributions

LH: Conceptualisation, Methodology, Software, Validation, Formal analysis, Data Curation, Writing - Original Draft; Visualisation
LGS.: Conceptualisation, Methodology, Software, Validation, Formal analysis, Data Curation, Writing - Original Draft; Visualisation
MMdB: Conceptualisation, Writing - Review & Editing;
GCG : Conceptualisation, Data Curation, Writing - Review & Editing
AMR : Conceptualisation, Data Curation, Writing - Review & Editing
DNB : Supervision, Funding acquisition, Writing - Review & Editing
EM : Supervision, Writing - Review & Editing

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JW : Conceptualisation
TMNC: Conceptualisation, Data Curation, Supervision, Writing - Review & Editing.

Competing Interests

The authors declare no competing interests.

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